

Build advanced and highly reliable industrial Ethernet communication system



ISM5X10D Managed Industrial Ethernet Switch

· Public Utility

· Smart Manufacturing

· Smart City

· Smart Energy

1. Product Overview

The ISM series of managed industrial Ethernet switches is specifically designed to meet the demands of harsh industrial environments such as power, transportation, and industrial control applications.

Product Features:

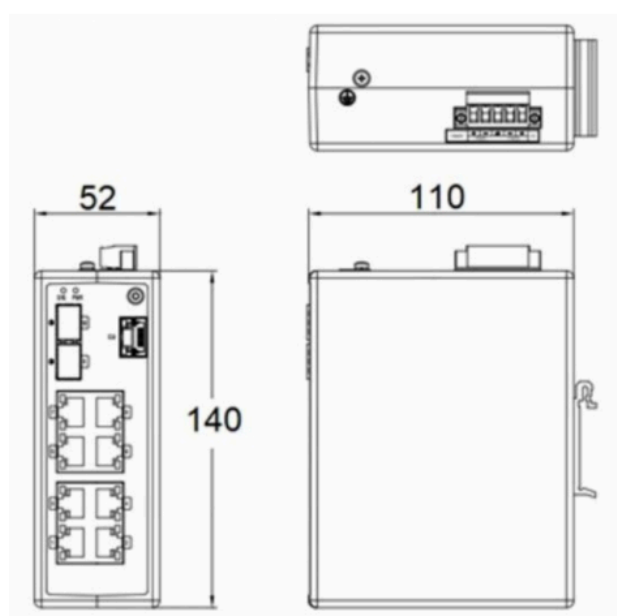
- **Reliability & Resilience:** Fanless design, IP30/IP40 protection rating, dust and dirt resistance, and wide temperature operation.
- **Redundancy Protocols:** Supports STP/RSTP/MSTP ring redundancy protocols, offering flexible choices for intricate industrial networks.
- **Network Management:** Supports SNMP for integrated network management and RMON for effective network monitoring and fault prediction.
- **Enhanced Security & Control:** Port-based VLAN, IEEE 802.1Q VLAN, GVRP, IGMP Snooping, and QoS to process critical data.
- **PoE Capabilities:** Supports IEEE 802.3af/at PSE, providing up to 30W output power per port (ISM5310D).

Key Technical Specifications

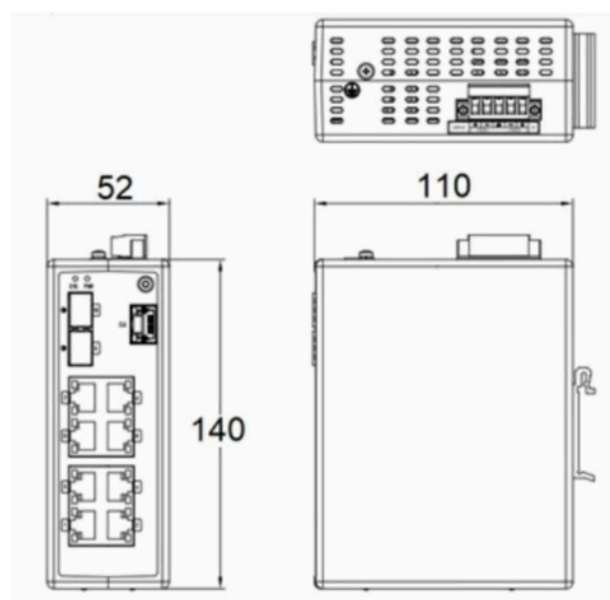
Parameter	Specification
Type	Layer 2 Managed Industrial Ethernet Switch
Ports	2 x 100/1000/2500BaseX SFP + 8 x 10/100/1000BaseT (ISM5010D); 8 x PoE 802.3af/at ports (ISM5310D)
Switching Performance	44 Gbps backplane bandwidth; 8K MAC table
Dimensions / Weight	52 mm x 140 mm x 110 mm; 0.7 kg
Power	18~60 VDC dual redundant (ISM5010D); 48~57 VDC dual redundant (ISM5310D)

Parameter	Specification
Environment	-40 to +75 °C operating; IP40/IP30
Management	Web, CLI, SNMPv1/v2c/v3, RMON; STP/RSTP/MSTP, IGMP Snooping, VLAN
Certifications	CE, FCC

2. Product Dimensions



ISM5010D



ISM5310D

Note:

1. All dimensions are in millimeters (mm).
2. All dimensions are approximate and **for reference only**.
3. The dimensions shown in the figure **shall not be used for production or processing**.
4. Dimensions must comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

3. Technical Specifications

3.1 Protocol Compliance List

Category/Parameter	Specification
IEEE Standards	
IEEE 802.3	CSMA/CD method and physical Layer specifications
IEEE 802.1p	Priority Queuing
IEEE 802.1q	VLAN tagging
IEEE 802.1d	Spanning Tree Algorithm
IEEE 802.1w	Rapid Spanning Tree
IEEE 802.1s	Multiple Spanning Tree
IEEE 802.3ac	VLAN Tagging
IEEE 802.1x	Authentication
IEEE 802.3ad	Link Aggregation
IEEE 802.3x	Flow Control
IEEE 802.3	Ethernet
IEEE 802.3u	Fast Ethernet
IEEE 802.3z	Gigabit Ethernet
IEEE 802	Networks
RFC Standards	
RFC 768	UDP

Category/Parameter	Specification
RFC 791	IP
RFC 792	ICMP
RFC 793	TCP
RFC 826	ARP
RFC 854	Telnet Client & Server
RFC 862	Echo Protocol
RFC 863	Discard Protocol
RFC 867	Daytime Protocol
RFC 868	Time Protocol
RFC 1059, 1119	NTPv1/2
RFC 1191	Path MTU Discovery
RFC 1403	BGP OSPF Interaction
RFC 1542	Bootstrap Extensions & DHCP
RFC 1994	PPP Challenge Handshake Authentication Protocol
RFC 2068	HTTP
RFC 213	DHCP Server
RFC 2138	RADIUS
RFC 2139	RADIUS Accounting
RFC 2236	IGMPv2
RFC 2474	DiffServ Precedence
RFC 2597	DiffServ Assured Forwarding

Category/Parameter	Specification
RFC 2598	DiffServ Expedited Forwarding
RFC 2865	Remote Authentication Dial In User Service (RADIUS)
RFC 3046	DHCP Relay Agent Information Option
RFC 3222	Forwarding Information Base (FIB)
Other Protocols	
GMRP	GARP
GVRP	GARP
SSH2	Secure Shell 2
IGMP	Snooping
SNMPv3	Supported

3.2 Hardware Specifications

Category/Parameter	Specification
Physical Performance	
Enclosure	Fully enclosed seamless metal enclosure
Dimensions (W × D × H)	52 mm × 140 mm × 110 mm
Weight	0.7 kg
Mounting Method	DIN-rail mounting
Cooling Method	Fanless cooling
Ingress Protection	IP40 / IP30
Storage Temperature	-40 °C ~ +85 °C

Category/Parameter	Specification
Operating Temperature	-40 °C ~ +75 °C
Humidity	5 ~ 95% (non-condensing)
Hardware Performance	
Backplane Bandwidth	44 Gbps
Transmission Mode	Parallel Storage Forwarding
MAC Table Size	8K
RAM	128 MB
Packet Buffer Size	1.5 Mbits
Exchange Rate	148,800 pps/100M ports; 1,488,000 pps/1000M ports
Power Parameters	
Input Voltage	18~60 VDC dual redundant input (ISM5010D) 48~57 VDC dual redundant input (ISM5310D)
Power Consumption	10 W (ISM5010D)
Overload Current Protection	Supported
Reverse Polarity Protection	Supported
PoE (ISM5310D)	
PoE 10/100/1000 BaseT Ports	8 ports
Maximum Power	15.4W (IEEE 802.3af); 30W (IEEE 802.3at)

Category/Parameter	Specification
Electromagnetic Characteristics	
EMI	FCC 47 CFR Part 15 Class A; EN55022 Class A
EMS	IEC(EN)61000-4-2, Class 4 IEC(EN)61000-4-4, Class 4 IEC(EN)61000-4-5, Class 4 IEC(EN)61000-4-11, Class 4 IEC(EN)61000-4-12, Class 4
Mechanical Characteristics	
Shock	IEC60068-2-27
Freefall	IEC60068-2-31
Vibration	IEC60068-2-6
Certifications	
Certifications	CE, FCC (ISM5010D / ISM5310D)
Quality Assurance	
Warranty Period	5 years
MTBF	35 years

4. Software Specifications

Category/Parameter	Specification
Software Functions	
Redundancy	STP, MSTP, RSTP, Port Trunking
Management Mode	Browser, Serial Port, STD-17 MIB-II, STD-58 SMIV2, STD-59 RMON, STD-62 SNMPv3, SNMPv2c, SNMPv1
Time Synchronization	NTP
Diagnostic Mode	Indicator light, Journal File, RMON, Port Mirroring, TRAP

5. Ordering Guide

Model	Description
ISM5010D-P-2GSFP-8GT-24	10-port Layer2 managed Industrial Switch. 2 * 100/1000/2500BaseX SFP Ports (SFP module not included), 8 * 10/100/1000BaseT Ports. 1 Management Serial CLI Port, IP40 Protection Class, Operating Temperature from -40°C to +75°C. Isolated Dual 18-60VDC Power Inputs.
ISM5310D-P-2GSFP-8GT-48	10-port Layer2 managed Industrial Switch. 2 * 100/1000/2500BaseX SFP Ports (SFP module not included), 8 * 802.3af/at PoE 10/100/1000BaseT Ports. 1 Management Serial CLI Port, IP30 Protection Class, Operating Temperature from -40°C to +75°C. Dual 48-57VDC Power Inputs.

6. Contact Us

- Website: [InHand Networks](https://www.inhandnetworks.com)
- Copyright: ©InHand Networks All rights reserved